Capstone Project Proposal Report
(Individual Report)

Instructions:
This form is to be completed by each student doing Project registration to fulfill their senior design or capstone requirement. It must be completed and submitted to your Guide. Each student must complete this form individually.
This report is to be completed during the starting of the semester, while the project description report will be completed during end of the semester.

Guide Approval (initials/date):		

CAP4001– Capstone Project Proposal Report

Student Name	**Chintakayala Karthikeya**
Student Register Number	**23BCE9594**
Programme	**CSE AI/ML**
Semester/Year	**7th Semester**
Guide(s)	**Balusa Bhanu Chander**
Project Title	**An Explainable AI-Based Decision Support System for Dynamic Water Quality Index Prediction and Recommendation System**

Team Composition: Provide the information below for each member of the **project team**. Include **all** project team members, not just those in your discipline or those enrolled for Capstone project. Please also include yourself!

Reg. No	Name	Major	Specialization
23BCE9594	Chintakayala Karthikeya	CSE	AI/ML
23BCE7692	Ameer Roshan Shaik	CSE	AI/ML
23BCE9533	Shaik Abdul Samad	CSE	AI/ML
23BCE8722	Desam Pranay Kumar Reddy	CSE	AI/ML

Project and Task Description:

(a) Project Summary

Water quality assessment helps in providing the safe water, protecting the environment, and managing the water resources in an effective manner. Now the process of determining the Water Quality Index (WQI) is done manually by considering the number of water quality parameters. It is therefore not effective for large-scale implementations.

The current study aims to develop an Explainable Artificial Intelligence (XAI) based Decision Support System for predicting Water Quality Index (WQI) using an appropriate Machine Learning Algorithm. The project will adopt 2000 water quality instances retrieved from water monitoring stations as the target dataset. Water quality parameters that will be considered include Temperature, Dissolved Oxygen, pH, Conductivity, Biological Oxygen Demand (BOD), Nitrate + Nitrite , Fecal Coliform, and Total Coliform.

The project will calculate Water Quality Index using a published standard methodology applicable in the target dataset. The project will develop a number of machine learning models including Random Forest, XGBoost, and LightGBM. The best performing model will be interpreted using SHAP values to understand the relationship between the input parameters and the predicted WQI. The project will make recommendations on water quality based on the predicted WQI.

(b) Individual Role and Tasks

As an individual member of project team, I will primarily contribute to the **machine learning and deep learning model development stages** of the project. My responsibilities include:

- Conduct literature review on the Water Quality Index prediction using machine learning and deep learning.

- Perform exploratory data analysis to understand relationship between water-quality parameters and WQI.
- Prepare feature sets and target variables for the predictive modelling.
- Develop multiple machine-learning regression models for the WQI prediction.
- Implementing and evaluate models such as Random Forest, Extra Trees, Gradient Boosting, XGBoost, and CatBoost.
- Develop deep-learning models including an Artificial Neural Network and the TabNet for WQI prediction.
- Apply a consistent experimental procedure for the both baseline and imputed datasets.
- Evaluate model performance using the MAE, RMSE, and R^2 score.
- Compare the performance of the machine-learning and deep-learning models.
- Perform hyperparameter tuning for the selected high-performing models.
- Prepare model-performance tables, graphs, and the comparative analysis.
- Contribute to the research paper writing, project documentation, and final presentation.

Individual Deliverables

Literature Review Contribution
Exploratory Data Analysis
Feature and Target Dataset Preparation
Random Forest Model
Extra Trees Model
Gradient Boosting Model
XGBoost Model
CatBoost Model
Artificial Neural Network
TabNet Model
Hyperparameter Tuning Results
Model Performance Comparison Report
Performance Visualization
Research Paper Contribution
Final Project Documentation
Presentation Slides

(c) Specific Approach

The given project will adopt the standard approach used in machine learning. First of all, the water quality data set will be collected and preprocessed. It includes removing missing values and other irrelevant information to prepare the data for training.

Then, with the use of the published method for calculating the Water Quality Index (WQI), that is applicable to the given data set, the WQI will be determined. Following that, the Exploratory Data Analysis will be performed to examine the relationship between various parameters and detect patterns in the data set.

After that, feature engineering will be applied, and various algorithms, among which Random Forest, XGBoost, and Ligther GBM, will be trained on the given data set. The performance of all the models will be compared using evaluation metrics such as Mean Absolute Error, Root Mean Square Error, and R2 Score. Then, the performance of the best model will be explained using the SHAP (SHapley Additive exPlanations). In the end, a Decision Support System will be designed that will indicate the predicted WQI and give a recommended statement on the quality of water based on the prediction.

(d) Design Phases

Phase Activities

Phase 1 Literature Review and Problem Identification
Phase 2 Dataset Collection and Data Preprocessing
Phase 3 Water Quality Index Calculation
Phase 4 Exploratory Data Analysis and Feature Engineering
Phase 5 Machine Learning Model Development
Phase 6 Model Training, Hyperparameter Tuning and Evaluation
Phase 7 Explainable AI Implementation using SHAP
Phase 8 Decision Support System Development
Phase 9 Testing, Documentation, Research Paper Preparation and Final Presentation

Outcome Matrix: Describe your plan to demonstrate each of the outcomes below.

Outcomes:	**Plan for demonstrating outcome:**
a) an ability to apply knowledge of mathematics, science, and engineering	Apply machine learning, deep learning, statistics, optimization, and data-analysis concepts to develop models for predicting Water Quality Index from water-quality parameters.
c) an ability to design a system, component, or process to meet desired needs within realistic constraints such as economic, environmental, social, political, ethical, health and safety, manufacturability, and sustainability	Design predictive modelling workflows while considering dataset size, computational requirements, model complexity, generalization performance, prediction accuracy, and available time and computing resources.
d) an ability to function on multidisciplinary teams	Collaborate with team members by using the prepared WQI datasets for modelling and providing prediction outputs required for explainability and decision-support development.
e) an ability to identify, formulate, and solve engineering problems	Formulate WQI prediction as a supervised regression problem and solve it using multiple machine-learning and deep-learning algorithms with systematic performance comparison.
g) an ability to communicate effectively	Present model architecture, experimental methodology, evaluation results, performance comparisons, tables, graphs, research paper content, and technical findings clearly.
k) an ability to use the techniques, skills, and modern engineering tools necessary for engineering practice	Use Python, VS Code, Pandas, NumPy, Scikit-learn, XGBoost, CatBoost, deep-learning frameworks, and visualization libraries for model development and evaluation.

Realistic Constraints:

There are various realistic constraints taken into consideration in the project including lack or imperfection of water quality readings, variance in water quality at various monitoring stations, computing constraints during model training, and availability of certain water quality parameters only for calculation of the Water Quality Index. Environmental sustainability, proper use of open-source data and the generation of reliable recommendations is also taken into account.

Engineering Standards:

The project will comply with the established engineering and research standards throughout its implementation. Standard machine learning techniques like data pre-processing, feature engineering, validation of models, hyperparameter tuning and evaluation of the models with MAE, RMSE and R^2 Score will be used. Explainable Artificial Intelligence (XAI) will be utilized via SHAP. The Decision Support System will provide recommendations on the basis of WQI values according to certain water quality categories. Documentation and research reporting will be done according to IEEE standards.